ITA06020 MACHINE LEARNING

Experiment 7 – Implementation of Logistic Regression

Aim:

To implement the Logistic Regression algorithm in Python for classification and predict the output using a real-world dataset.

Dataset: Breast Cancer Wisconsin Dataset (available in Scikit-learn).

Procedure

1. Import the required Python libraries.
2. Load the Breast Cancer dataset.
3. Separate input features and target values.
4. Split the dataset into training and testing data.
5. Normalize the input features.
6. Create a Logistic Regression model.
7. Train the model using training data.
8. Predict the output and calculate accuracy.

Python Program

```
from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

data = load_breast_cancer()

X = data.data

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42

)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Logistic Regression")

print("Predicted Output:", data.target_names[y_pred[0]])

print("Actual Output:", data.target_names[y_test[0]])

print("Accuracy:", round(accuracy_score(y_test, y_pred)*100, 2), "%")
```

Output:

```
Logistic Regression
Predicted Output: benign
Actual Output: benign
Accuracy: 97.37 %
```

Result

Thus, the Logistic Regression algorithm was successfully implemented using Python and the Breast Cancer dataset. The model achieved an accuracy of 97.37%.